



# Cours d'introduction : bases d'astronomie

*M1 ASTRO - CCP HAP702P Astrophysique*

*Bertrand Plez*



# Outline

---

- 1 Enseignement d'astrophysique en Master Astrophysique/CCP
- 2 Astrophysical objects and orders of magnitude
- 3 Positional astronomy
- 4 Bibliographic ressources

# Outline

---

- 1 Enseignement d'astrophysique en Master Astrophysique/CCP
- 2 Astrophysical objects and orders of magnitude
- 3 Positional astronomy
- 4 Bibliographic ressources

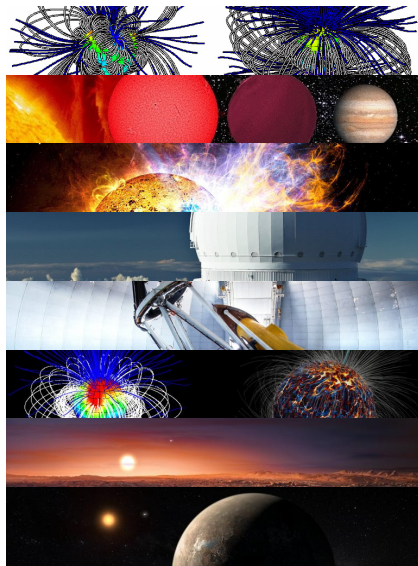
# Présentations

## ■ Bertrand Plez

- [bertrand.plez@umontpellier.fr](mailto:bertrand.plez@umontpellier.fr)

## ■ Professeur

- Faculté des Sciences
  - Physique
  - Astrophysique
- Laboratoire Univers et Particules de Montpellier
  - Atmosphères stellaires



# Enseignement d'astrophysique en Master Astrophysique/CCP

## ■ M1

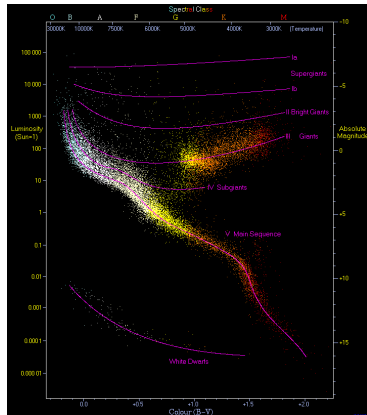
- HAP702P Astrophysique
- HAP703P Atelier Astrophysique Observationnelle 1

## ■ M2 Astrophysique (et CCP)

- HAP903P Transfert Radiatif et Atmosphères Stellaires
- HAP916P Milieu Interstellaire
- HAP905P Atelier Astrophysique Observationnelle 2 (OHP)

## ■ M2 Astrophysique

- HAP903P Structure et Évolution Stellaire
- HAP005P Projet Num. Astro.
- HAP912P9 Form. Etoiles Syst. Planét.
- HAP913P9 Form. Evol. Galaxies
- HAP910P9 Cosmo Observationnelle
- HAP002P9 Instru. Astrophysique



*Diagramme de Hertzsprung-Russell*

*Crédit: R. Powell*

# Enseignement d'astrophysique en Master

## Astrophysique/CCP

### ■ M1

- HAP702P Astrophysique
- HAP703P Atelier Astrophysique Observationnelle 1

### ■ M2 Astrophysique (et CCP)

- HAP903P Transfert Radiatif et Atmosphères Stellaires
- HAP916P Milieu Interstellaire
- HAP905P Atelier Astrophysique Observationnelle 2 (OHP)

### ■ M2 Astrophysique

- HAP903P Structure et Évolution Stellaire
- HAP005P Projet Num. Astro.
- HAP912P9 Form. Etoiles Syst. Planét.
- HAP913P9 Form. Evol. Galaxies
- HAP910P9 Cosmo Observationnelle
- HAP002P9 Instru. Astrophysique



OHP/T152 Crédit : Ministère de la Culture



M15 Crédit : D. Minenna/V. Niederhorn/R. Ruffault

# Déroulement M1 HAP702P

---

## ■ HAP702P : 24h en séances cours et/ou TD 1,5h ou 3h

- Objets astrophysiques et astronomie de position
- Rayonnement EM et son étude
- Processus de rayonnement et photométrie
- Propriétés observationnelles des étoiles
- Atmosphères stellaires et transfert de rayonnement
- Spectroscopie astrophysique
- Structure stellaire
- Formation stellaire

## ■ Évaluation

- 2 sessions : 100% contrôle terminal

## ■ <https://moodle.umontpellier.fr/>

- [cours HAP702P](#) (Tou.te.s. inscrit.e.s?)

# Outline

---

1 Enseignement d'astrophysique en Master Astrophysique/CCP

2 Astrophysical objects and orders of magnitude

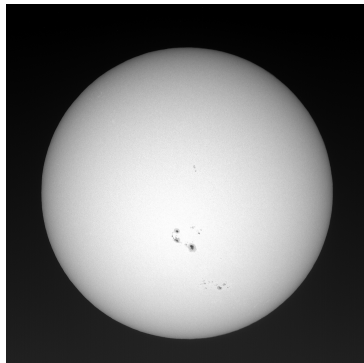
- The solar system
- Stars and exoplanets
- The Galaxy : the Milky Way

3 Positional astronomy

4 Bibliographic ressources

# The solar system : the Sun (1/4)

- $M_{\odot} = 1.988 \times 10^{30} \text{ kg}$
- $R_{\odot} = 6.957 \times 10^8 \text{ m}$
- $L_{\odot} = 3.828 \times 10^{26} \text{ W}$
- $T_{\text{eff},\odot} = 5,772 \text{ K}$
- âge: 4.57 Ga
- Composition :
  - mass: H: 73,81 %    He: 24,85 %  
+ 1,34 % “metals”  
*Asplund et al. (2009)*  
O, C, Ne, N, Fe, Mg, Si, S , and all other species
  - number: H: 90,97 %    He: 8,89 %  
+ 0,14 % “metals”

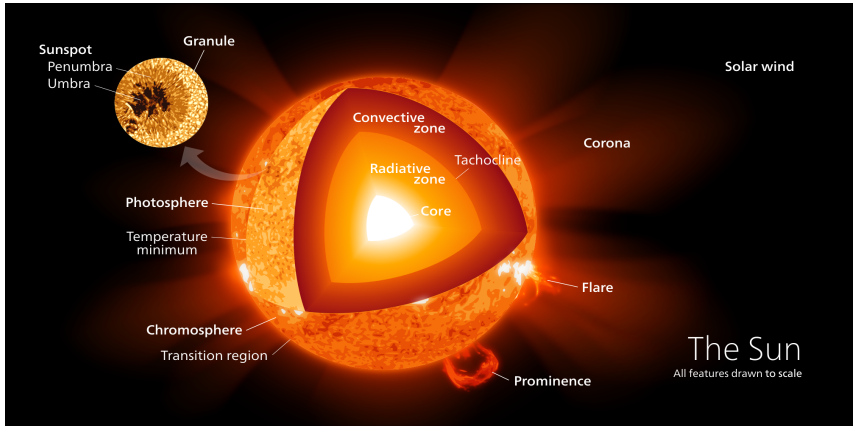


*The sun at visible wavelength  
H. Reboul Univ. Montpellier*

→ [NASA Sun fact sheet](#)

→ [IAU resolution 2015 B3](#)

# The solar system : the Sun (2/4)



# The solar system : the Sun (3/4)

## ■ Rotation

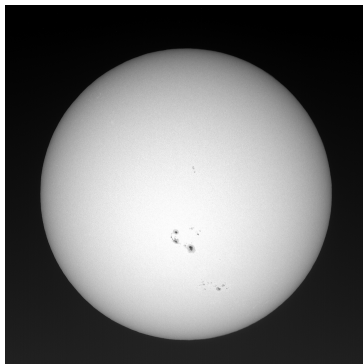
- poles: 32 d
- equator: 24 d

## ■ Magnetic field

- sunspots :  $0.2 - 0.4 \text{ T}$  ( $2 - 4 \text{ kG}$ )
- dipole field :  $1 - 2 \times 10^{-4} \text{ T}$  ( $1 - 2 \text{ G}$ )
- local field :  $1 - 2 \times 10^{-2} \text{ T}$  ( $100 - 200 \text{ G}$ )
- NB: Earth:  $30 \text{ to } 60 \mu\text{T}$

## ■ Extended atmosphere

- chromosphere
  - $T$  up to  $30,000 \text{ K}$
  - thickness:  $2,500 \text{ km}$
- corona
  - $T \sim 1 - 2 \text{ MK}$  (up to  $30 \text{ MK}$  in flares)
  - extension: several  $R_{\odot}$



*The Sun at visible wavelength  
H. Reboul Univ. Montpellier*

# The solar system : the Sun (3/4)

## ■ Rotation

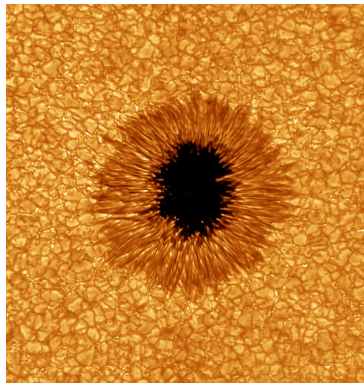
- poles: 32 d
- equator: 24 d

## ■ Magnetic field

- sunspots :  $0.2 - 0.4 \text{ T}$  ( $2 - 4 \text{ kG}$ )
- dipole field :  $1 - 2 \times 10^{-4} \text{ T}$  ( $1 - 2 \text{ G}$ )
- local field :  $1 - 2 \times 10^{-2} \text{ T}$  ( $100 - 200 \text{ G}$ )
- *NB: Earth: 30 to 60  $\mu\text{T}$*

## ■ Extended atmosphere

- chromosphere
  - $T$  up to 30,000 K
  - thickness: 2,500 km
- corona
  - $T \sim 1 - 2 \text{ MK}$  (up to 30 MK in flares)
  - extension: several  $R_{\odot}$



*AR 1084, TiO (706 nm) filter, AO  
Big Bear Observatory*

# The solar system : the Sun (3/4)

## ■ Rotation

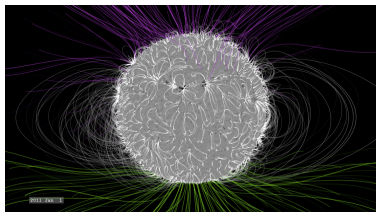
- poles: 32 d
- equator: 24 d

## ■ Magnetic field

- sunspots :  $0.2 - 0.4 \text{ T}$  ( $2 - 4 \text{ kG}$ )
- dipole field :  $1 - 2 \times 10^{-4} \text{ T}$  ( $1 - 2 \text{ G}$ )
- local field :  $1 - 2 \times 10^{-2} \text{ T}$  ( $100 - 200 \text{ G}$ )
- NB: Earth:  $30 \text{ to } 60 \mu\text{T}$

## ■ Extended atmosphere

- chromosphere
  - $T$  up to  $30,000 \text{ K}$
  - thickness:  $2,500 \text{ km}$
- corona
  - $T \sim 1 - 2 \text{ MK}$  (up to  $30 \text{ MK}$  in flares)
  - extension: several  $R_{\odot}$



*Global solar magnetic field (PFSS)  
January 2011*

*Credit: NASA/SVS*

# The solar system : the Sun (3/4)

## ■ Rotation

- poles: 32 d
- equator: 24 d

## ■ Magnetic field

- sunspots :  $0.2 - 0.4 \text{ T}$  ( $2 - 4 \text{ kG}$ )
- dipole field :  $1 - 2 \times 10^{-4} \text{ T}$  ( $1 - 2 \text{ G}$ )
- local field :  $1 - 2 \times 10^{-2} \text{ T}$  ( $100 - 200 \text{ G}$ )
- *NB: Earth: 30 to 60  $\mu\text{T}$*

## ■ Extended atmosphere

- chromosphere
  - $T$  up to 30,000 K
  - thickness: 2,500 km
- corona
  - $T \sim 1 - 2 \text{ MK}$  (up to 30 MK in flares)
  - extension: several  $R_{\odot}$

*Solar granulation*  
*TiO (706 nm) filter*  
*Big Bear Observatory*  
*FoV=20,300 × 18,125 km*

# The solar system : the Sun (3/4)

## ■ Rotation

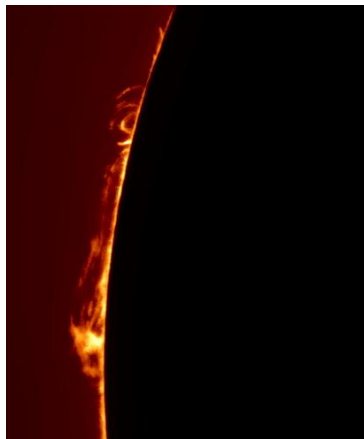
- poles: 32 d
- equator: 24 d

## ■ Magnetic field

- sunspots :  $0.2 - 0.4 \text{ T}$  ( $2 - 4 \text{ kG}$ )
- dipole field :  $1 - 2 \times 10^{-4} \text{ T}$  ( $1 - 2 \text{ G}$ )
- local field :  $1 - 2 \times 10^{-2} \text{ T}$  ( $100 - 200 \text{ G}$ )
- NB: Earth:  $30 \text{ to } 60 \mu\text{T}$

## ■ Extended atmosphere

- chromosphere
  - $T$  up to  $30,000 \text{ K}$
  - thickness:  $2,500 \text{ km}$
- corona
  - $T \sim 1 - 2 \text{ MK}$  (up to  $30 \text{ MK}$  in flares)
  - extension: several  $R_{\odot}$



*The solar chromosphere  
at  $656.3 \text{ nm}$  ( $H\alpha$ )*

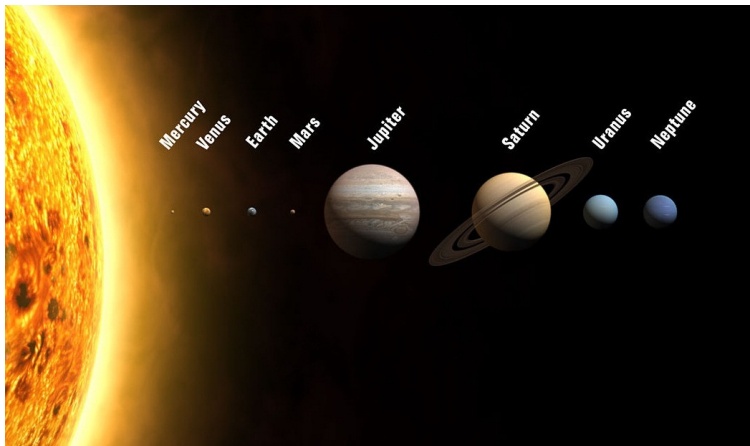
# The solar system : the Sun (4/4)

---

*SOHO, ESA/NASA*

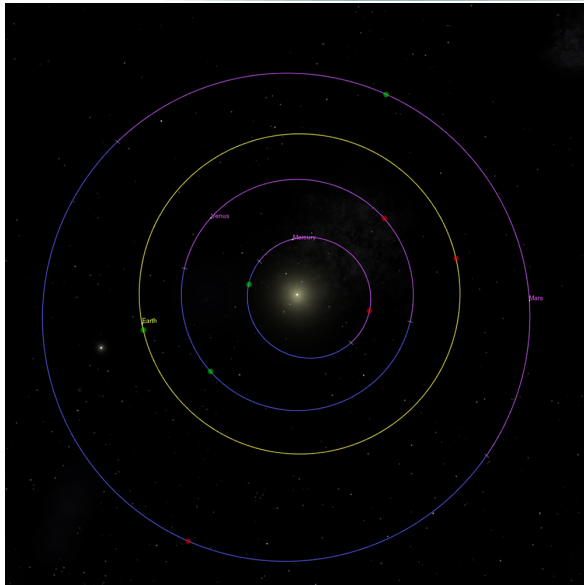
# The solar system : planets (1/4)

---



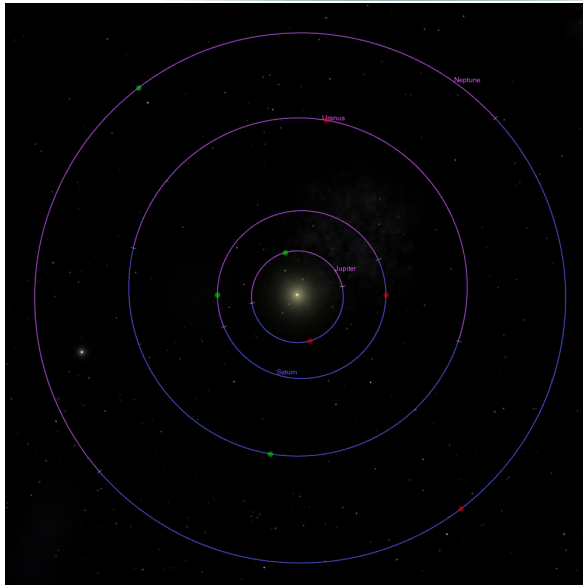
*Planet diameters to scale*

# The solar system : planets (1/4)



*Inner solar system: orbits to scale*

# The solar system : planets (1/4)



*Outer solar system: orbits to scale*

# The solar system : planets (2/4)

|                                       | MERCURY | VENUS   | EARTH  | MOON   | MARS  | JUPITER  | SATURN   | URANUS   | NEPTUNE  | PLUTO   |
|---------------------------------------|---------|---------|--------|--------|-------|----------|----------|----------|----------|---------|
| <b>Mass</b> ( $10^{24}$ kg)           | 0.330   | 4.87    | 5.97   | 0.073  | 0.642 | 1898     | 568      | 86.8     | 102      | 0.0146  |
| <b>Diameter</b> (km)                  | 4879    | 12,104  | 12,756 | 3475   | 6792  | 142,984  | 120,536  | 51,118   | 49,528   | 2370    |
| <b>Density</b> (kg/m <sup>3</sup> )   | 5427    | 5243    | 5514   | 3340   | 3933  | 1326     | 687      | 1271     | 1638     | 2095    |
| <b>Gravity</b> (m/s <sup>2</sup> )    | 3.7     | 8.9     | 9.8    | 1.6    | 3.7   | 23.1     | 9.0      | 8.7      | 11.0     | 0.7     |
| <b>Escape Velocity</b> (km/s)         | 4.3     | 10.4    | 11.2   | 2.4    | 5.0   | 59.5     | 35.5     | 21.3     | 23.5     | 1.3     |
| <b>Rotation Period</b> (hours)        | 1407.6  | -5832.5 | 23.9   | 655.7  | 24.6  | 9.9      | 10.7     | -17.2    | 16.1     | -153.3  |
| <b>Length of Day</b> (hours)          | 4222.6  | 2802.0  | 24.0   | 708.7  | 24.7  | 9.9      | 10.7     | 17.2     | 16.1     | 153.3   |
| <b>Distance from Sun</b> ( $10^6$ km) | 57.9    | 108.2   | 149.6  | 0.384* | 227.9 | 778.6    | 1433.5   | 2872.5   | 4495.1   | 5906.4  |
| <b>Perihelion</b> ( $10^6$ km)        | 46.0    | 107.5   | 147.1  | 0.363* | 206.6 | 740.5    | 1352.6   | 2741.3   | 4444.5   | 4436.8  |
| <b>Aphelion</b> ( $10^6$ km)          | 69.8    | 108.9   | 152.1  | 0.406* | 249.2 | 816.6    | 1514.5   | 3003.6   | 4545.7   | 7375.9  |
| <b>Orbital Period</b> (days)          | 88.0    | 224.7   | 365.2  | 27.3   | 687.0 | 4331     | 10,747   | 30,589   | 59,800   | 90,560  |
| <b>Orbital Velocity</b> (km/s)        | 47.4    | 35.0    | 29.8   | 1.0    | 24.1  | 13.1     | 9.7      | 6.8      | 5.4      | 4.7     |
| <b>Orbital Inclination</b> (degrees)  | 7.0     | 3.4     | 0.0    | 5.1    | 1.9   | 1.3      | 2.5      | 0.8      | 1.8      | 17.2    |
| <b>Orbital Eccentricity</b>           | 0.205   | 0.007   | 0.017  | 0.055  | 0.094 | 0.049    | 0.057    | 0.046    | 0.011    | 0.244   |
| <b>Axial Tilt</b> (degrees)           | 0.01    | 177.4   | 23.4   | 6.7    | 25.2  | 3.1      | 26.7     | 97.8     | 28.3     | 122.5   |
| <b>Mean Temperature</b> (C)           | 167     | 464     | 15     | -20    | -65   | -110     | -140     | -195     | -200     | -225    |
| <b>Surface Pressure</b> (bars)        | 0       | 92      | 1      | 0      | 0.01  | Unknown* | Unknown* | Unknown* | Unknown* | 0       |
| <b>Number of Moons</b>                | 0       | 0       | 1      | 0      | 2     | 67       | 62       | 27       | 14       | 5       |
| <b>Ring System?</b>                   | No      | No      | No     | No     | No    | Yes      | Yes      | Yes      | Yes      | No      |
| <b>Global Magnetic Field?</b>         | Yes     | No      | Yes    | No     | No    | Yes      | Yes      | Yes      | Yes      | Unknown |
|                                       | MERCURY | VENUS   | EARTH  | MOON   | MARS  | JUPITER  | SATURN   | URANUS   | NEPTUNE  | PLUTO   |

## NASA Planetary fact sheet (metric)

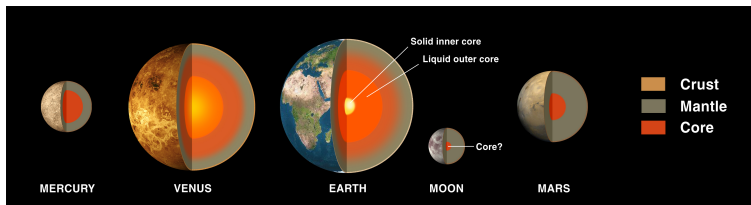
# The solar system : planets (2/4)

|                        | MERCURY | VENUS  | EARTH | MOON     | MARS  | JUPITER  | SATURN   | URANUS   | NEPTUNE  | PLUTO   |
|------------------------|---------|--------|-------|----------|-------|----------|----------|----------|----------|---------|
| Mass                   | 0.0553  | 0.815  | 1     | 0.0123   | 0.107 | 317.8    | 95.2     | 14.5     | 17.1     | 0.0025  |
| Diameter               | 0.383   | 0.949  | 1     | 0.2724   | 0.532 | 11.21    | 9.45     | 4.01     | 3.88     | 0.186   |
| Density                | 0.984   | 0.951  | 1     | 0.605    | 0.713 | 0.240    | 0.125    | 0.230    | 0.297    | 0.380   |
| Gravity                | 0.378   | 0.907  | 1     | 0.166    | 0.377 | 2.36     | 0.916    | 0.889    | 1.12     | 0.071   |
| Escape Velocity        | 0.384   | 0.926  | 1     | 0.213    | 0.450 | 5.32     | 3.17     | 1.90     | 2.10     | 0.116   |
| Rotation Period        | 58.8    | -244   | 1     | 27.4     | 1.03  | 0.415    | 0.445    | -0.720   | 0.673    | 6.41    |
| Length of Day          | 175.9   | 116.8  | 1     | 29.5     | 1.03  | 0.414    | 0.444    | 0.718    | 0.671    | 6.39    |
| Distance from Sun      | 0.387   | 0.723  | 1     | 0.00257* | 1.52  | 5.20     | 9.58     | 19.20    | 30.05    | 39.48   |
| Perihelion             | 0.313   | 0.731  | 1     | 0.00247* | 1.41  | 5.03     | 9.20     | 18.64    | 30.22    | 30.16   |
| Aphelion               | 0.459   | 0.716  | 1     | 0.00267* | 1.64  | 5.37     | 9.96     | 19.75    | 29.89    | 48.49   |
| Orbital Period         | 0.241   | 0.615  | 1     | 0.0748   | 1.88  | 11.9     | 29.4     | 83.7     | 163.7    | 247.9   |
| Orbital Velocity       | 1.59    | 1.18   | 1     | 0.0343   | 0.808 | 0.439    | 0.325    | 0.228    | 0.182    | 0.157   |
| Orbital Eccentricity   | 12.3    | 0.401  | 1     | 3.29     | 5.60  | 2.93     | 3.38     | 2.74     | 0.677    | 14.6    |
| Axial Tilt             | 0.0004  | 0.113* | 1     | 0.285    | 1.07  | 0.134    | 1.14     | 4.17*    | 1.21     | 2.45*   |
| Surface Pressure       | 0       | 92     | 1     | 0        | 0.01  | Unknown* | Unknown* | Unknown* | Unknown* | 0       |
| Number of Moons        | 0       | 0      | 1     | 0        | 2     | 67       | 62       | 27       | 14       | 5       |
| Ring System?           | No      | No     | No    | No       | No    | Yes      | Yes      | Yes      | Yes      | No      |
| Global Magnetic Field? | Yes     | No     | Yes   | No       | No    | Yes      | Yes      | Yes      | Yes      | Unknown |
|                        | MERCURY | VENUS  | EARTH | MOON     | MARS  | JUPITER  | SATURN   | URANUS   | NEPTUNE  | PLUTO   |

## NASA Planetary fact sheet (relative to Earth)

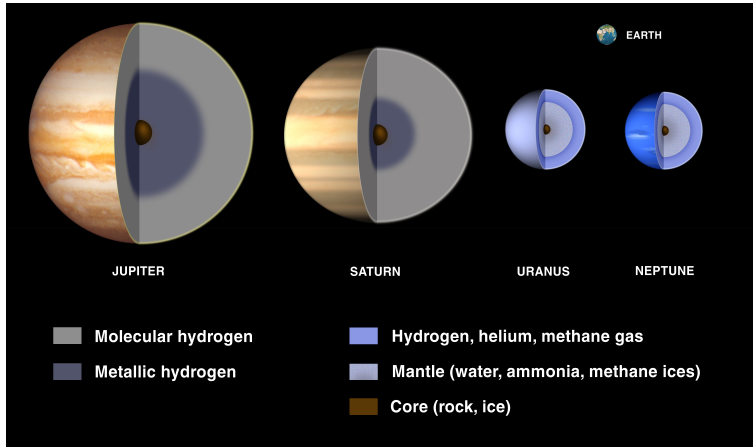
- Relevant unit for planetary orbits
  - astronomical unit:  $1 \text{ au} = 149,597,870,700 \text{ m}$

# The solar system : planets (3/4)



NASA

# The solar system : planets (3/4)



NASA

# The solar system : planets (4/4)

## ■ What is a planet?

### ➔ IAU resolution B5-B6 (2006)

- (1) A planet<sup>1</sup> is a celestial body that
  - (a) is in orbit around the Sun,
  - (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape,<sup>2</sup> and
  - (c) has cleared the neighbourhood around its orbit.
- (2) A "dwarf planet" is a celestial body that
  - (a) is in orbit around the Sun,
  - (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape<sup>2</sup>,
  - (c) has not cleared the neighbourhood around its orbit, and
  - (d) is not a satellite.
- (3) All other objects<sup>3</sup>, except satellites, orbiting the Sun shall be referred to collectively as "Small Solar System Bodies".

---

<sup>1</sup> The eight planets are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.

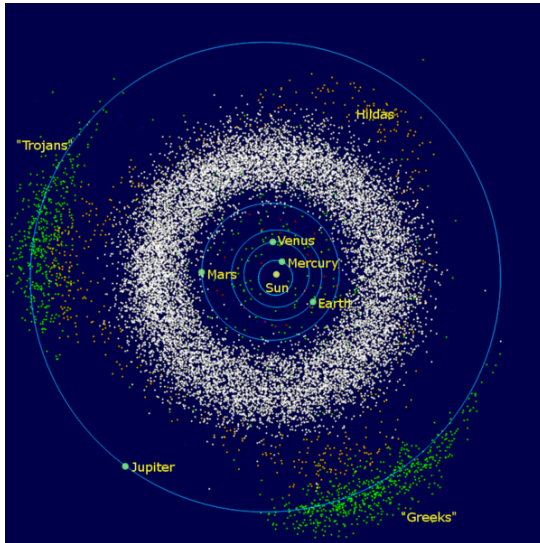
<sup>2</sup> An IAU process will be established to assign borderline objects to the dwarf planet or to another category.

<sup>3</sup> These currently include most of the Solar System asteroids, most Trans-Neptunian Objects (TNOs), comets, and other small bodies.

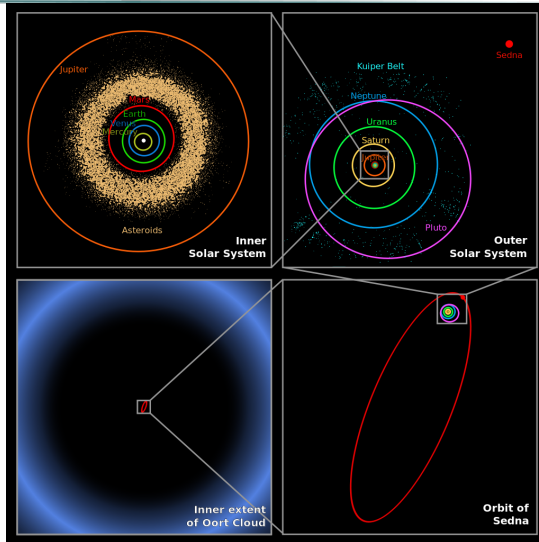
Pluto is a "dwarf planet" by the above definition and is recognized as the prototype of a new category of Trans-Neptunian Objects<sup>1</sup>.

# The solar system : dwarf planets, TNOs (trans-neptunian objects), asteroids and comets

---

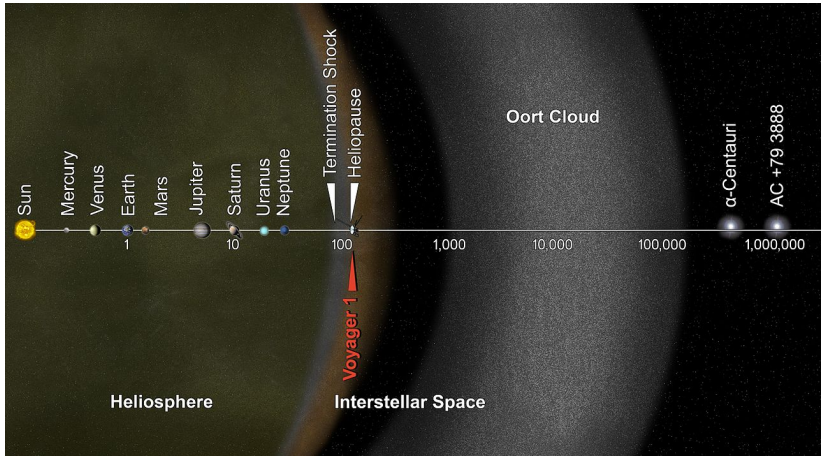


# The solar system : dwarf planets, TNOs (trans-neptunian objects), asteroids and comets



NASA

# The solar system : dwarf planets, TNOs (trans-neptunian objects), asteroids and comets



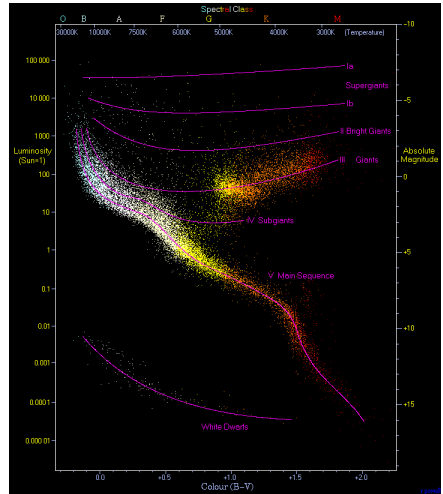
NASA

*au logarithmic scale*

# Stars : the Hertzsprung-Russell diagram

■  $L_{\star} = f(T_{\text{eff}})$

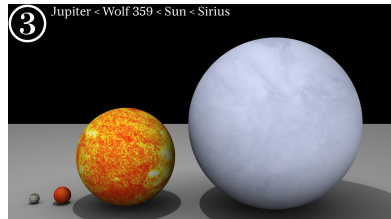
- Most stars on two sequences
  - Main sequence
  - Giant branch(es)
- Fewer stars in two other groups
  - White dwarfs
  - Supergiants



# Stars : main sequence parameter range

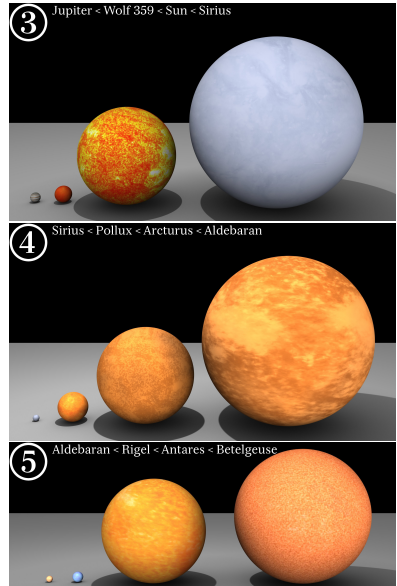
---

- Mass:  
 $0.08\text{--}100\ M_{\odot}$
- Radius:  
 $0.1\text{--}20\ R_{\odot}$
- Temperature:  
 $2,500\text{--}40,000\ \text{K}$
- Luminosity:  
 $10^{-3}\text{--}10^6\ L_{\odot}$



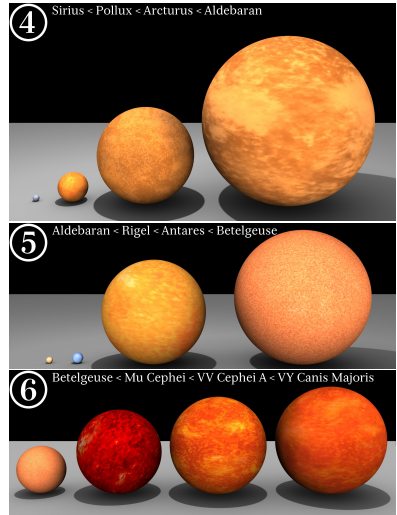
# Stars : full parameter range

- Radius range:  
 $10^{-2}$ – $1\,000\,R_{\odot}$
- Not visible on HRD
  - neutron stars
  - black holes
  - stellar remnant/compact objects
    - ↳ like white dwarfs



# Stars : full parameter range

- Radius range:  
 $10^{-2}$ – $1\,000\,R_{\odot}$
- Not visible on HRD
  - neutron stars
  - black holes
- ➔ stellar remnant/compact objects
  - ➡ like white dwarfs



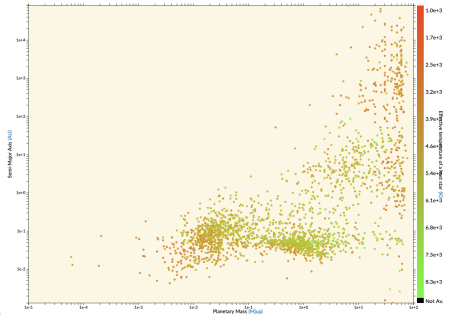
# Planetary systems

6, 486 known planets in 4, 862 systems,  
1, 078 multiple planet systems  
(23/09-2026)

- Systems very  $\neq$  from solar
- Variety of host stars
- Conditions on these planets?

See press releases

- Nobel Prize 2019 for the discovery of the first exoplanet: 51 Peg b
- Hot jupiters in stellar nurseries?
- Magnetic Field and Wind of Kappa Cet: Toward the Planetary Habitability of the Young Sun When Life Arose on Earth
- A terrestrial planet candidate in a temperate orbit around Proxima Cen
- Conférence Proxima b FdS



Source: *exoplanet.eu*

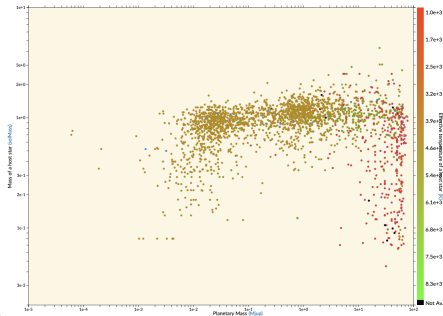
# Planetary systems

6, 486 known planets in 4, 862 systems,  
1, 078 multiple planet systems  
(23/09-2026)

- Systems very  $\neq$  from solar
- Variety of host stars
- Conditions on these planets?

See press releases

- Nobel Prize 2019 for the discovery of the first exoplanet: 51 Peg b
- Hot jupiters in stellar nurseries?
- Magnetic Field and Wind of Kappa Cet: Toward the Planetary Habitability of the Young Sun When Life Arose on Earth
- A terrestrial planet candidate in a temperate orbit around Proxima Cen
- Conférence Proxima b FdS



Source: *exoplanet.eu*

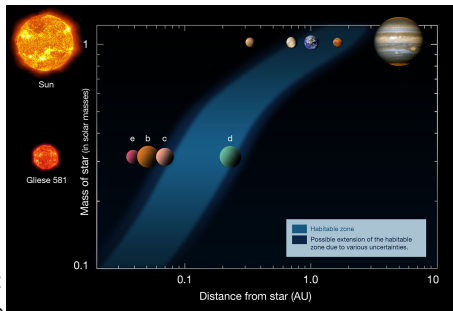
# Planetary systems

6, 486 known planets in 4, 862 systems,  
1, 078 multiple planet systems  
(23/09-2026)

- Systems very  $\neq$  from solar
- Variety of host stars
- Conditions on these planets?

See press releases

- Nobel Prize 2019 for the discovery of the first exoplanet: 51 Peg b
- Hot jupiters in stellar nurseries?
- Magnetic Field and Wind of Kappa Cen: Toward the Planetary Habitability of the Young Sun When Life Arose on Earth
- A terrestrial planet candidate in a temperate orbit around Proxima Cen
- Conférence Proxima b FdS



*ESO*

# Planets or brown dwarfs?

- Low-mass star to brown dwarf transition
  - Stars produce energy through H-fusion and have a MS phase
  - $M < 0.08 M_{\odot}$ : no H-fusion: brown dwarfs
- Where is the brown dwarf to planet transition ?
  - Energy source (IAU working definition)
    - $M > 65 M_J$ : lithium fusion
    - $M > 13 M_J$ : deuterium fusion → minimum BD mass
  - Formation process
    - accretion in a circumstellar disc → planet
    - star-like collapse → BD
- Continuum of objects
- Constantly increasing number of known BD and giant planets



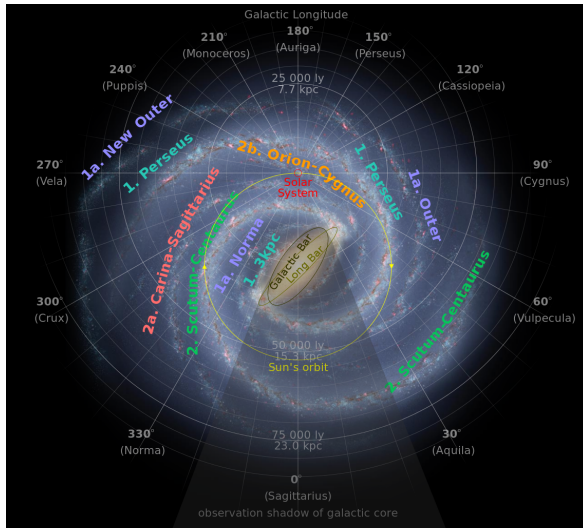
# The Galaxy : the Milky Way

---



*The Milky Way as seen from Earth – equator-on*

# The Galaxy : the Milky Way



*Artist impression of the Milky Way as seen pole-on*

*Credit: NASA*

# The Milky Way: figures

---

## ■ Suitable units

- Light-year:  $1 \text{ ly} = 9.461 \times 10^{15} \text{ m} = 6.324 \times 10^4 \text{ au}$   
IAU def: the distance that light travels in vacuum in one Julian year
- Parsec:  $1 \text{ pc} = 3.086 \times 10^{16} \text{ m} = 2.063 \times 10^5 \text{ au} = 3.262 \text{ ly}$

## ■ Dimensions: diameter: 30 kpc; thickness: 300 pc

## ■ Total mass: $10^{12} M_{\odot}$ / $\sim 80 - 90 \%$ dark matter

## ■ Composition (ordinary matter):

- stars (85 – 90 %;  $2 - 4 \times 10^{11}$  stars)
- interstellar medium (10 – 15 %; gas and dust)

# Outline

---

1 Enseignement d'astrophysique en Master Astrophysique/CCP

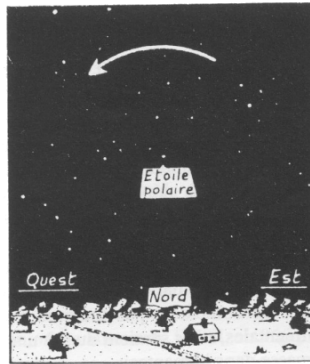
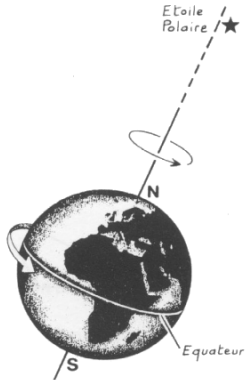
2 Astrophysical objects and orders of magnitude

3 Positional astronomy

- Back to basics
- Azimuthal coordinates
- Equatorial coordinates
- Sidereal time and hour angle
- Object Observability

4 Bibliographic ressources

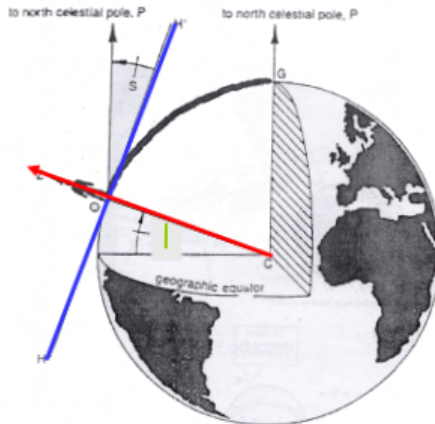
# Back to basics



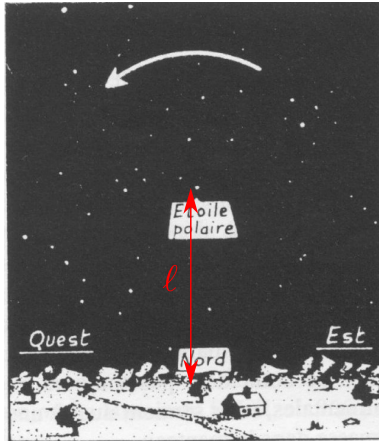
©Acker (Ganeval)

A. Acker

# Back to basics



# Back to basics



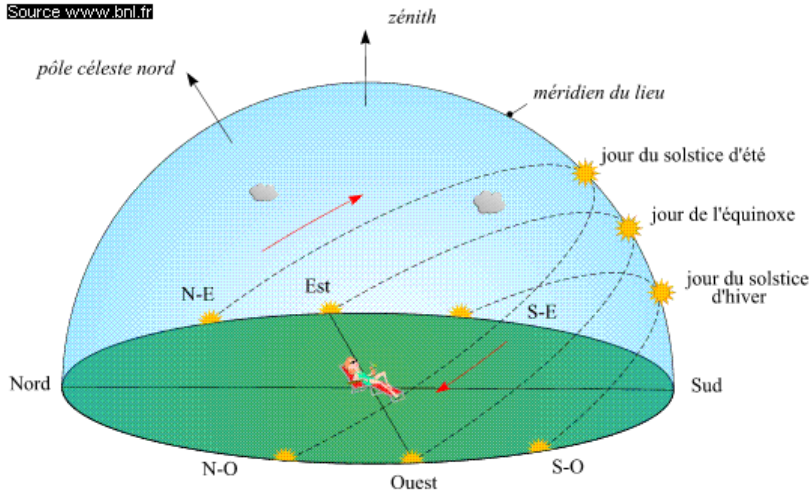
©Acker (Ganeval)

# Back to basics

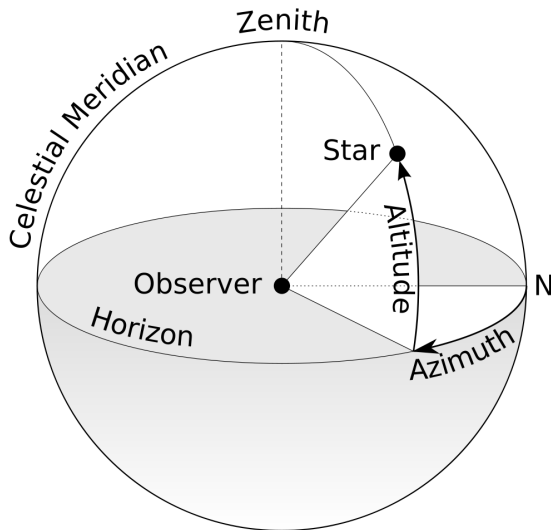
---

# Back to basics

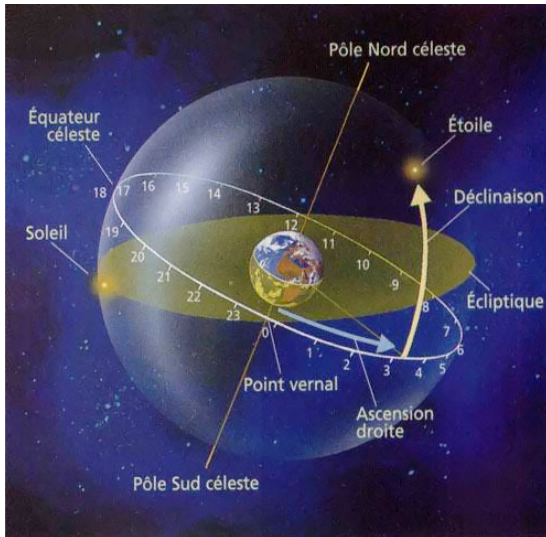
Source [www.bnl.fr](http://www.bnl.fr)



# Azimuthal coordinates



# Equatorial coordinates

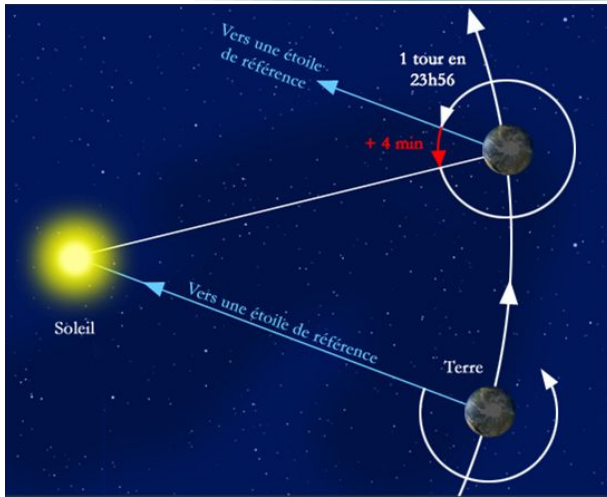


# Equatorial coordinates

---

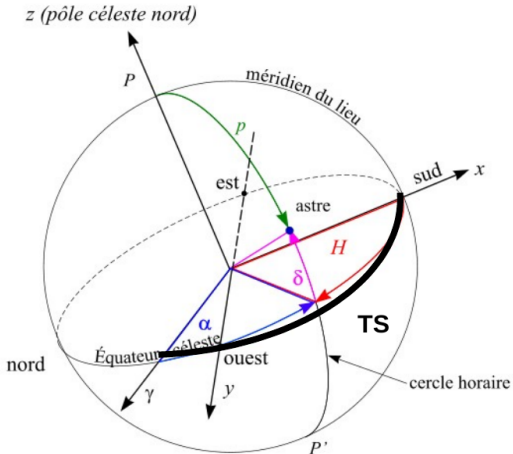
- Declination: DEC /  $\delta$  : angle between celestial equator and object
  - Analog to latitude for geographic coordinates
- Right Ascension: RA /  $\alpha$  : angle between vernal equinox  $\gamma$  and object
  - Analog to longitude for geographic coordinates
  - The line  $\alpha = 0\text{h}$  rotates for an observer on Earth
- Example/notation :
  - Betelgeuse ( $\alpha$  Ori) :  $\alpha = 05\text{h}55\text{m}10\text{s}$  ;  $\delta = +07^\circ 24' 25''$

# Sidereal time and hour angle



- Solar/calendar day :  $24\text{h}00\text{m}00\text{s} = 86\,400\text{ s}$
- Sidereal day :  $23\text{h}56\text{m}04\text{s} = 86\,164\text{ s}$  (approx. value)

# Sidereal time and hour angle



■ Experiment with Stellarium !

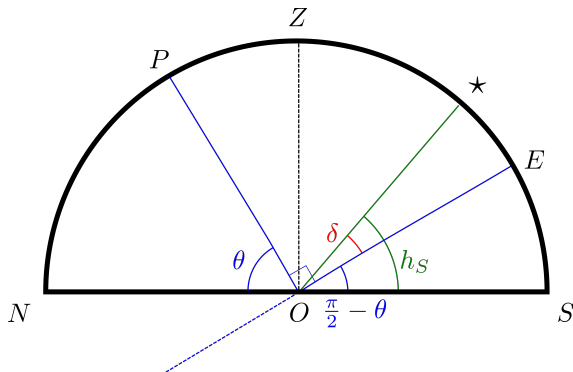
# Sidereal time and hour angle

- LST (Local Sidereal Time) : angle between South and  $\gamma$ 
  - ➡ LST =  $\alpha$  of objects crossing the meridian
- HA (Hour Angle) : between South and star's projection on celestial equator
  - ➡ HA = -1h ➡ : star will cross meridian in 1h
  - ➡ TS = HA of  $\gamma$

$$HA + \alpha = LST$$

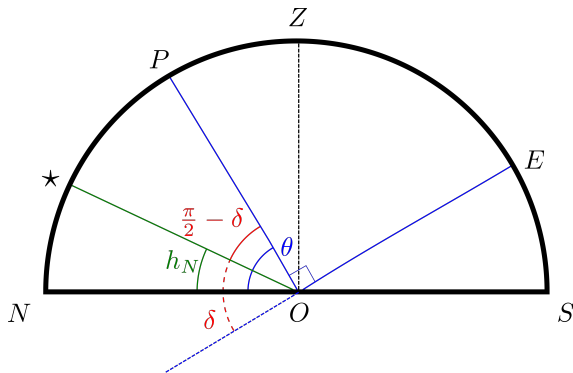
- ST depends on observer's longitude
  - LST = GST - longitude
  - GST : Greenwich Sidereal Time
- LST(midnight) =  $\alpha_{\odot} + 12\text{h}$ 
  - Spring equinox :  $\alpha_{\odot} = 0\text{h} \Rightarrow \text{LST}(\text{midnight}) = 12\text{h}$
  - Summer solstice :  $\alpha_{\odot} = 6\text{h} \Rightarrow \text{LST}(\text{midnight}) = 18\text{h}$
  - ...
  - LST shifts by 4min a day, 2h a month

# Maximum elevation above horizon



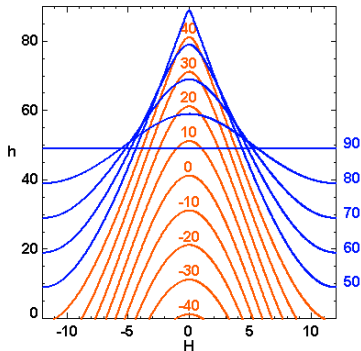
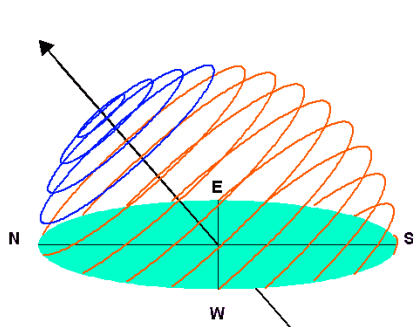
- Northern hemisphere, latitude  $\theta > 0$  [Montpellier :  $\theta_M = 43^\circ 37'$ ]
  - Elevation at (southern) meridian crossing
    - $h_{\max} = 90 - \theta + \delta$
  - Visible above horizon:  $h_{\max} > 0^\circ$  :  $\delta > \theta - 90$  [Mpl :  $\delta > -46^\circ 23'$ ]
  - under good conditions :  $h_{\max} > 30^\circ$  :  $\delta > \theta - 60$  [Mpl :  $\delta > -16^\circ 23'$ ]

# Circumpolar objects



- Northern hemisphere, latitude  $\theta > 0$  [Montpellier :  $\theta_M = 43^\circ 37'$ ]
  - Elevation at (northern) meridian crossing
    - $h_{\min} = \theta - (90 - \delta) = \delta + \theta - 90$
  - Visible above horizon:  $\delta > 90 - \theta$  [Mpl :  $\delta > 46^\circ 23'$ ]
  - under good conditions :  $h_{\min} > 30^\circ$  :  $\delta > 120 - \theta$  [Mpl :  $\delta > 76^\circ 23'$ ]

# Hour angles at rise and set



*Credit : Astrophysique sur Mesure – Observatoire de Paris*

■ Rise :  $HA_{\ell} = -\arccos(-\tan \theta \cdot \tan \delta)$

■ Set :  $HA_c = +\arccos(-\tan \theta \cdot \tan \delta)$

# Outline

---

- 1 Enseignement d'astrophysique en Master Astrophysique/CCP
- 2 Astrophysical objects and orders of magnitude
- 3 Positional astronomy
- 4 Bibliographic ressources**

- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod  
ISBN-13: 978-2100576401
  - Chapitre 3 : "Positions et mouvements des astres"
  - Chapitre 5 : "Analyser une étoile : le Soleil"
  - Chapitre 9 : "Les systèmes planétaires"
  - Chapitre 10 : "Notre galaxie : - La Voie Lactée"
- *"Les étoiles et le Milieu Interstellaire"*, R. Monier, 2006, Ellipses  
ISBN-13: 978-2729827861
- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences  
ISBN-13: 978-2868838773
- *"The hunt for Vulcan"*, T. Levenson et al., 2016, Random House Trade  
ISBN-13: 978-0812988307